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Section: A (POWER)
Assignment: 01
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Write a single C++ statement to accomplish each of the following:

- i. Declare the variables voltage, current, and resistance to be of type double.

```
double voltage, current, resistance;
```

```
// Create three decimal-number variables to store voltage, current, and resistance
```

- ii. Display the message “Welcome to Object Oriented Programming in C++!” on the screen.

```
cout << "Welcome to Object Oriented Programming in C++!";
```

```
// Show a welcome message on the screen for the user
```

iii. Read a value from the user and store it in an integer variable named age.

```
cin >> age;
```

```
// Take age input from the keyboard and store it inside variable age
```

iv. If the variable temperature is greater than 100, print “Overheating detected!”.

```
if (temperature > 100) cout << "Overheating detected!";
```

```
// Check temperature; if it crosses 100, warn about overheating
```

v. Compute the area of a circle with radius r and store it in area ($\pi = 3.14159$).

```
area = 3.14159 * r * r;
```

```
// Calculate circle area using  $\pi r^2$  and save the result in area
```

vi. Print the values of x, y, and z separated by a comma.

```
cout << x << ", " << y << ", " << z;
```

```
// Print x, y, and z neatly separated by commas
```

vii. Assign the sum of voltage and current to total.

```
total = voltage + current;
```

```
// Add voltage and current, then store the result in total
```

viii. Display "Error: Division by zero" only if denominator equals zero.

```
if (denominator == 0) cout << "Error: Division by zero";  
  
// Show error message only when denominator becomes zero
```

ix. Increment the variable count by 1 and then print its new value.

```
cout << ++count;  
  
// Increase count by 1 first, then print the updated value
```

x. Read three floating-point numbers from the keyboard into a, b, and c.

```
cin >> a >> b >> c;  
  
// Take three decimal numbers from user and store them in a, b, and c
```

xi. Total resistance in series (take values from user).

```
cin >> R1 >> R2 >> R3, cout << "Total Resistance (Series) = " << (R1 + R2 + R3);  
  
// Take three resistor values and add them because series resistances simply add
```

xii. Total resistance in parallel (take values from user).

```
cin >> R1 >> R2 >> R3, cout << "Total Resistance (Parallel) = " << 1 / ((1/R1) + (1/R2) + (1/R3));  
  
// Take three resistances and apply parallel resistance formula
```

xiii. Voltage across R2 using Voltage Division Rule.

(Chosen values: $R1 = 10\Omega$, $R2 = 20\Omega$, $V = 12V$)

```
cout << "Voltage across R2 = " << 12 * (20.0 / (10 + 20));
```

```
// Use voltage division rule to calculate voltage across R2
```

xiv. Current through R2 using Current Division Rule.

(Chosen values: $R1 = 10\Omega$, $R2 = 20\Omega$, $V = 12V$)

```
cout << "Current through R2 = " << (12.0 / (10 + 20)) * (10.0 / (10 + 20));
```

```
// Find total current then apply current division rule to get current through R2
```

xv. Calculate Semester GPA.

```
cin >> credit >> grade, totalCredits += credit, totalPoints += credit * grade;
```

```
// Enter credit hours and grade points, then update totals
```

```
cout << "Semester GPA = " << (totalPoints / totalCredits);
```

```
// Divide total grade points by total credit hours to calculate GPA
```